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Proof-of-principle demonstration of endogenous circadian system and circadian misalignment effects on human oral microbiota

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Abstract

Circadian misalignment—the misalignment between the central circadian clock and behavioral and environmental cycles (including sleep/wake, fasting/eating, dark/light)—results in adverse metabolic effects. Potential underlying mechanisms for these adverse effects include alterations in orogastrointestinal microbiota. However, it remains unknown whether human oral microbiota has endogenous circadian rhythmicity and whether circadian misalignment influences oral microbiota community composition. Healthy young individuals [27.3 ± 2.3 years (18–35 years),

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Declaration of Interests

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2 women, body-mass index range: 18–28 kg/m^2] were enrolled in a stringently controlled 14-day laboratory circadian protocol. This included a 32h constant routine (CR) protocol (circadian baseline assessment), a Forced Desynchrony (FD) protocol with four 28-h “days” under ~3 lux to induce circadian misalignment, and a post-misalignment 40h CR protocol. Microbiota assessments were performed on saliva samples collected every 4h throughout both CR protocols. Total DNA was extracted and processed using high-throughput 16S ribosomal RNA gene amplicon sequencing. The relative abundance of specific oral microbiota populations, i.e., one of the five dominant phyla, and three of the fourteen dominant genera, exhibited significant endogenous circadian rhythmicity. Importantly, circadian misalignment dramatically altered the oral microbiota landscape, such that four of five dominant phyla and eight of fourteen dominant genera exhibited significant circadian misalignment effects. Moreover, circadian misalignment significantly affected metagenome functional content of the oral microbiota (inferred gene content analysis), as revealed by changes in specific functional pathways associated with metabolic control and immunity. Collectively, our proof-of-concept study provides evidence for human oral microbiota circadian rhythmicity and show that even relatively short-term experimental misalignment can dramatically modify microbial population structure and functional pathways involved in metabolism and immunity. These findings have translational relevance to individuals typically exposed to circadian misalignment, as night shift workers and frequent flyers.

Keywords

Circadian rhythms; circadian misalignment; shift work; microbiota; functional pathways

Introduction

Night shift work is highly prevalent in the United States (1) and Europe (2), and is a risk factor for type 2 diabetes mellitus (T2DM), obesity, and cardiovascular disease (3–5). Shift workers typically experience circadian misalignment, which is misalignment between the central circadian clock and behavioral and environmental cycles (sleep/wake, fasting/eating, rest/activity, dark/light) (6). Another physiological condition that involves circadian misalignment is jet lag, which is caused by rapid long-distance trans-meridian (east–west or west–east) travel (7). Circadian misalignment impairs glucose tolerance, decreases insulin sensitivity (8–11), and increases inflammatory factors and blood pressure (12–14). Potential mechanisms underlying impaired glucose control and immune activation may include microbiota alterations that influence the orogastrointestinal barrier integrity and immune system function (15–17). Animal experiments have shown that both environmental and genetic circadian rhythm disruption perturb microbial communities and can cause intestinal hyperpermeability (18–21). Importantly, mice with disrupted circadian rhythmicity show exaggerated proinflammatory responses to lipopolysaccharide (a component in the outer membrane of Gram-negative bacteria) exposure and greater mortality (22). Collectively, these animal data highlight the adverse consequences of the disruption of circadian rhythms on microbiota communities.

Currently, diurnal - not circadian - rhythms have been shown for gut microbiota during sleep/wake and fasting/feeding cycles in animal models (21, 23) and for oral microbiota in

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humans (17, 24, 25). Furthermore, there is some evidence for diurnal (morning vs. evening) differences in oral and rectal microbiome in humans assessed under masked conditions (26). However, from these studies, it cannot be determined if these variations are the consequence of the eating/fasting and wake/sleep cycles, or a result of an endogenous circadian rhythm, thus persistent in the absence of the rhythms in food intake, sleep, physical activity, and light exposure (which may have strong effects themselves on the microbiota).

Emerging evidence shows a pivotal role of the orogastrointestinal microbiota in the regulation of many physiological processes in humans (*e.g.,* digestion of food components, host metabolism, immune system) (27, 28). Yet, it remains unknown whether human microbiota populations exhibit endogenous circadian rhythms and whether circadian misalignment affects the structure and function of microbial communities. In this study, we focused on oral microbiota, which have clinical relevance (e.g., periodontitis increasing cardiovascular disease risk (29), and—critically—allows for non-invasive, high-temporal resolution sampling in humans that is required for assessing circadian rhythms and which is not possible with stool microbiota collections in humans. Frequent stool microbiota sampling is not only invasive, but also does not allow high-temporal resolution because it is an aggregate of microbiota over many hours or even days. Here we addressed the two-fold question of: (1) whether human oral microbial communities have an endogenous circadian rhythm independent of the fasting/eating, sleep/wake, and dark/light cycle; and (2) whether circadian disruption affects microbial community structure. This study was performed using a 14-day stringently controlled circadian laboratory protocol in healthy humans. We demonstrate that the relative abundance of specific human oral bacterial taxa exhibits endogenous circadian rhythmicity, i.e., persisting in the absence of 24-h rhythms in behavioral and environmental cycles. Importantly, we find proof-of-concept evidence that circadian misalignment changes the relative abundance of a wide array of oral microbial taxa and specific functional pathways.

Methods and Materials

Participants

The protocol was approved by the institutional review board at the Partners Human Research Committee, performed in accordance with the principles of the Declaration of Helsinki, and participants provided written informed consent. Six healthy young individuals [27.3 ± 2.3 years (18–35 years), 2 women, body-mass index (BMI) range: 18–28 kg/m^2] were enrolled in the laboratory protocol. Participants were free from any medical conditions, and medical suitability was determined by medical history, biochemical and toxicology screenings of blood and urine, and physical and psychological exams. Participants were free from medication intake, smoking and drug (except for oral contraceptives). Children were excluded given the intensive nature of the laboratory protocols and the upper age limit excluded older individuals with higher prevalence of co-morbidities and medication intake and to enhance homogeneity. One woman was admitted on menstrual cycle days 1–5, and one during days 14–19.

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Study design

Participants maintained a fixed, self-selected habitual bedtime for two weeks before the laboratory with 8-h time in bed. Compliance was verified with ambulatory actigraphy (Actiwatch, Respironics), sleep logs, and time-stamped voicemails. During the three days before the laboratory study, participants received all meals prepared by the Dietary Core to meet dietary requirements, based on Harris Benedict formula with activity factor of 1.4 and controlled macronutrient distribution (25% fat, 60% carbohydrate, and 15% protein), which matched the in-laboratory protocol. Following the ambulatory phase, participants were admitted to the 14-day stringently-controlled circadian laboratory protocol (Figure 1). During the laboratory protocol, participants remained in individual laboratory suites, in a time-free environment, but maintained regular contact with staff members. The study protocol had two baseline days, followed by a 32h constant routine (CR) protocol, a gold-standard procedure to assess endogenous circadian rhythmicity (circadian baseline assessment). During the CR, participants remained awake, in semi-recumbent posture, with light intensity ~3lx, and hourly isocaloric snacks (30). Subsequently, they had a 16h sleep opportunity (thereby representing a 2:1 ratio of schedule wakefulness to scheduled sleep), and two recovery days with additional sleep opportunity (naps), to recover from the CR protocol and dissipate homeostatic sleep pressure. On baseline days (days 1–2) and on recovery CR days (days 5–6), participants had two short wake episodes (~1h) in dim light conditions *(3 lx)* during their scheduled sleep to perform e.g., visual analogue scales during the scheduled short wakefulness in the FD protocol. On day 7, participants underwent a Forced Desynchrony (FD) protocol (four 28-h “days” under ~3lx; outside the range of circadian entrainment (31) to induce circadian misalignment, such that by the 4th FD day participants were 12h out of phase (misaligned) relative to the 1st FD day (circadian aligned). During each 28h ‘‘day,’’ the ratio of the duration of scheduled awakening (18h:40min) and sleep (9h:20min) was maintained at 2:1, to match the baseline days. Participants had standardized meals at 0h:10min, 4h:10min, 8h:10min, and 12h:10min since scheduled awakening, and had to consume meals within 20min. After four FD cycles, participants underwent a post-misalignment 40h CR protocol, 16h sleep recovery opportunity, and were then discharged.

Diet

During the CR protocol, participants received an isocaloric diet (i.e., CR snacks) calculated according to the Harris-Benedict equation (32) with an activity factor of 1.2. Diet consisted of 45–50% carbohydrate, 15–20% protein, 30–35% fat, with 150 mEq Na+ (± 20%), 100 mEq K+ (±20%), and at least 2.5 L of water per 24 h. CR snacks comprised two alternating CR options based on a food preference form for each participantś two CR preselected CR snack choices (2 of 6 snack choices with different ingredients but same macronutrient composition). CR snacks were calculated with the same two snack options and same caloric level throughout the both CR protocols.

During the FD protocol, participants received meals (i.e., breakfast, lunch, snack and dinner) standardized across days based on a food preference form per participant. Diet was calculated according to the Harris-Benedict equation (32) with an activity factor of 1.4, and consisted of 45–50% carbohydrate, 15–20% protein, 30–35% fat, with 150 mEq Na+ (±

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20%), 100 mEq K+ (±20%), and at least 2.5 L of water per 24 h. The energy content of the meals/snack (% of total "day's" calorie intake) was Breakfast: 33.3% (± 35 kcals); Lunch: 23.4% (± 20 kcals); Snack: 10% (± 10 kcals); Dinner: 33.3% (± 35 kcals).

During the other laboratory segments, participants received an isocaloric diet calculated according to the Harris–Benedict equation (32) with an activity factor of 1.4. The diet had the same macronutrient composition as for the CR and FD segments. Participants were instructed to consume all food provided (verified by checking their food trays).

Central circadian phase assessment

Central circadian phase assessment was determined from core body temperature (CBT). Participants had a flexible rectal temperature sensor (Yellow Springs Instrument Company, OH, USA) and they used it continuously throughout the laboratory study (except during showers and bowel movements) for CBT measurement. Non-orthogonal cosinor analyses of each participant's CBT data were applied to estimate circadian phase (31). The circadian period used included a range from 23.4h to 24.7h (33). Average period estimation across participants and including both CR protocols was 24.49 ± 0.54h, with no significant differences between the baseline and post-misalignment CR protocols. Fitted circadian CBT minimum was assigned as the reference phase marker of 0° for each individual, and all microbiota data were then assigned a specific circadian phase according to collection time (0°−360°).

Saliva sample collection

Prior to collecting the first saliva sample, a staff member reviewed a standardized saliva collection procedure with the participant, which included restrictions of oral care/hygiene and food intake. During the CR protocols, oral care/hygiene included only mouthwash (without alcohol) once at the start of CR. No gum chewing was allowed; no food or water was provided in the 20 minutes prior to the saliva sample. One saliva sample was collected at each designated time per participant, using a sterile tube. Each saliva sample had a minimum of 2 mL of 'un-stimulated' saliva. Participants were instructed to wear gloves whilst producing a saliva sample.

DNA Extraction and 16S rRNA Gene Amplicon Sequencing Analysis

Oral salivary microbiota data were generated from six participants during two circadian protocols. Baseline CR1 (before misalignment) consisted of seven time points (24 hours total) with 4h intervals between collection time points. CR2 (after misalignment) consisted of nine time points (32 hours total) with 4hr intervals between collection time points. The first saliva samples in both CR protocols included in these analyses commenced ~5h:20min after scheduled awakening to avoid potential masking effects of changes in posture, light exposure and behavioral state (30). For each saliva sample, we collected a minimum of 2 mL of unstimulated saliva. Total DNA was extracted from saliva samples utilizing the FastDNA bead-beating Spin Kit for Soil (MP Biomedicals, Solon, OH, USA), and verified with fluorometric quantitation (Qubit 3.0, Life Technologies, Grand Island, NY). Primers 515FB/806RB (515FB:GTGYCAGCMGCCGCGGTAA; 806RB:GGACTACNVGGGTWTCTAAT) targeting the V4 variable region of microbial small subunit (SSU or 16S) ribosomal

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RNA (rRNA) genes were used for PCR (34), and prepared for high-throughput amplicon sequencing using a modified two-step targeted amplicon sequencing (TAS) approach, as described previously (35). Negative controls were used with each set of amplifications, which indicated no contamination. Samples were pooled in equal volume using an EpMotion5075 liquid handling robot (Eppendorf, Hamburg, Germany). The pooled library was purified using an AMPure XP cleanup protocol (0.6X, vol/vol; Agencourt, Beckmann-Coulter) to remove fragments smaller than 300bp. The pooled libraries, with a 20% phiX spike-in, were loaded onto an Illumina MiniSeq flow cell (2×153 paired-end reads) and sequenced. Based on the distribution of reads per barcode, the amplicons (before purification) were re-pooled to generate a balanced distribution of reads. The re-pooled and re-purified libraries were again sequenced on a MiniSeq flow cell. Library preparation, pooling, and MiniSeq sequencing were performed at the Genome Research Core (GRC) at the University of Illinois at Chicago.

Raw sequence data (FASTQ files) for each sample were merged using the software package PEAR (Paired-end-read merger) (v0.9.8) (36). Merged reads were quality trimmed and sequences shorter than 250 bases were discarded (CLC Genomics Workbench, v10.0, CLC Bio, Qiagen, Boston, MA). Sequences were screened for chimeras (usearch8.1 algorithm) (37), and putative chimeric sequences were removed from the dataset (QIIME v1.8) (38). Each saliva sample was rarefied (37,000 sequences/sample) and data were pooled and clustered into operational taxonomic units (OTU) at 97% similarity (Usearch8.1 algorithm). Representative sequences from each OTU were extracted and classified using a consensus taxonomy assigner (Greengenes 13_8 reference database). A biological observation matrix (BIOM) (39) was generated at each taxonomic level from phylum to species (“make OTU table” and “summarize taxa” algorithm). Differences in the relative abundance of individual taxa (>1%) between circadian condition groups were assessed for significance with respect to circadian rhythmicity and as the impact of circadian misalignment on microbial community composition of the oral microbiota. Taxa with an average abundance of <1% were removed from the analysis. The software package PICRUSt (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States) was used to predict the abundance of functional Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways based on the 16S rRNA gene amplicon sequence data (40). PICRUSt abundance data was not rarefied, but rather filtered at 0.01% threshold. Any pathway that had an overall abundance of at least 0.01% of the total abundance was retained for analysis.

Statistical analysis

Primary statistical analysis examined the effects of the circadian cycle and circadian alignment conditions assessed by cosinor analyses using mixed model analyses of variance (PROC MIXED with SAS version 9.4 (SAS Institute, Cary, NC, USA). The cosinor analyses PROC MIXED model was applied to the microbiota relative abundance at the taxonomical levels of phylum and genus (for those with a relative abundance threshold >1%, thus, dominant phyla and genera). We applied cosinor mixed model analyses for biorhythmic data based on fitting a circadian rhythm with a series of sine and cosine curves (41). These analyses include autoregressive error structures to model the strong serial correlation often seen in biorhythmic data, and the use of mixed models to account for subject-to-subject

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biological variation (42). Our cosinor analyses used a pair of ~24-h sine and ~24-h cosine curves to identify a “circadian effect” (a fundamental circadian component of ~24-h). This combination of a cosine and sine curves could fit a rhythm with any phase. Furthermore, individual circadian period was based on the CBT period estimate, and a “linear effect” was modelled to account for any gradual changes across time into the CR.

Microbiota composition data were normalized based on the percentage of each participant’s average during baseline CR to minimize inter-individual differences. These cosinor mixed-models included main effect “circadian alignment condition” (baseline CR: circadian alignment or post-misalignment CR: circadian misalignment), “circadian effect”, “linear effect” and the interaction of “circadian alignment condition” and “circadian effect”. Participant was included as a random factor. The cosinor analyses included seven time point measurements during the baseline CR and nine time point measurements during the post-misalignment CR. Hence, the main factors “circadian alignment condition”, “circadian effect” and “linear effect” include 16 measurements across both CR protocols. Because of that, part of the e.g., circadian misalignment effects are due to time in the study protocol. We used cosinor mixed-models as primary analyses, as they allowed modeling circadian effects, circadian misalignment effects and their interaction (thus accounting for repeated measures), as well as linear effects, which assumes potential linear increases or decreases of a given variable (e.g., changes in relative abundance in oral microbiota community structures) across time, included in a single model. The benefit of this combined model is that this allows accounting for any linear trends across the CR protocol, e.g., due to the extended wakefulness throughout the protocol. Due to the sample size and the complexity of the cosinor mixed-model analyses applied, false discovery rate adjustments were not performed in this proof-of-principle study to avoid potential false negatives. Unless specified, data correspond to mean and standard error of mean. Inferred functional pathway differential abundances were determined using the software program STAMP (43). The STAMP analyses included the seven time point measurements during baseline CR and the nine time point measurements during post-misalignment CR. The STAMP analyses were performed by combining the baseline samples of all participants and comparing them to all post-misalignment samples from all participants (baseline CR: n = 42 vs. post-misalignment CR: n = 54). These analyses were performed to identify differentially abundant KEGG pathways using White’s non-parametric *t*-test. Significance for all statistical tests was set as $p<0.05$.

In addition to the cosinor mixed models analysis, we applied the ARSER (autoregressive spectral estimation regression) algorithm (44). We used the meta2d function within the MetaCycle package implemented within the R programming language (45), to detect circadian oscillations of the bacterial relative abundances of genera (˃1%) in a short time series in each of the six individuals, as well as grouping the six individuals together per CR conditions. The *p*-values (ARS_*p*value) and Benjamini-Hochberg corrected false discovery rate values (ARS_BH.Q), both indicated oscillation significance for each individual as well as for the group across the CR. Additionally, ARSER identified the phase and amplitude for each individual or group’s bacterial genera. The MetaCycle package used a period set at 24h. Of note, the cosinor mixed models analysis and ARSER may produce different outcomes. The cosinor mixed models analysis uses a linear approach

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based on more predictable rhythms, and ARSER algorithm uses a non-linear multiple variable approach for data that may have harmonic rhythms. Furthermore, the effect of prior circadian misalignment was based on the cosinor analyses that included the circadian alignment condition (baseline CR [which included seven time points] vs. post-misalignment CR [which included nine time points]), circadian effect (~24-h sine and ~24-h cosine curves) and a linear component. Therefore, the effects of prior circadian misalignment are adjusted for any differences in circadian phase and linear component. Alpha-diversity indices (i.e., Shannon, Simpson, richness, and Pielou's evenness) were calculated on rarefied data (37,000 sequences/samples) (46), using the package "vegan" implemented in the R programming language (https://cran.r-project.org, https://github.com/vegandevs/vegan). Differences in alpha-diversity indices were assessed using independent *t*-test, based on Shapiro-wilk test of normality. Significance levels were set at $p<0.05$. Assessment of global differences in microbial communities between individual's circadian conditions used pairwise Bray-Curtis dissimilarity (non-phylogenetic) metric, that was generated using the Primer7 software package (PRIMER-E Ltd., Lutton, United Kingdom), and used to perform analysis of similarity (ANOSIM). Significance of ANOSIM values were determined using 999 permutation calculations using log (x+1) analysis transformed data. Significance levels were set at $p<0.05$. Differential abundance of individual taxa (˃1%) between circadian conditions were assessed for significance using Kruskal-Wallis test controlling for false-discovery rate (FDR), implemented within the software package QIIME (47). The parameter log2 fold change was calculated for each individual according to the formula Log2FC = log2 (meanCR2/meanCR1).

Materials Availability and Data and Code Availability Statement

Raw sequence data (FASTQ files) were deposited in the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA), under the BioProject identifier PRJNA529227.

Results

Oral microbiota exhibits endogenous circadian rhythms

During Baseline CR, we detected significant endogenous circadian rhythms in the relative abundance of one of the five dominant phyla, and in three of the fourteen dominant genera. Bacteria from the phylum Proteobacteria expressed significant circadian rhythmicity ("circadian effect": F=5.1; $p=0.03$). Bacterial genera with significant circadian oscillations, included *Neisseria* (phylum Proteobacteria; "circadian effect": F=9.6; $p=0.003$), *Fusobacterium* (phylum Fusobacteria; "circadian effect": F=6.2; $p=0.04$) and *Streptococcus* (phylum Firmicutes; "circadian effect": F=6.1; $p=0.04$) (Figures 2A-D).

MetaCycle's ARSER analysis of bacterial genera grouped by CR also confirmed significant oscillations of *Neisseria*, along with *Capnocytophaga, Gemellaceae, Prevotella, and Lautropia* (ARSER results for each individual CR condition are shown in Table S1). Of note, there were no significant circadian effects for alpha-diversity indices when using Metacycle's ARSER analyses (data not shown).

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Circadian misalignment affects the relative abundance of oral microbiota populations

We observed that four of five dominant phyla and eight of fourteen dominant genera exhibited significant circadian misalignment effects (using cosinor mixed-model analyses), indicating a widespread effect on the relative abundances of the oral microbiota populations. Significant misalignment effects were detected for the more prevalent phyla, Firmicutes (Prevalence: 48.1%; Effect of "circadian alignment condition": F=15.3; p=0.002) and Proteobacteria (Prevalence: 23.1%; Effect of "circadian alignment condition": F=5.9; p=0.017). Significant misalignment effects were also observed for less prevalent phyla, Bacteroidetes (Prevalence: 14.7%; Effect of "circadian alignment condition": F =11.2; p=0.001) and Fusobacteria (Prevalence: 6.9%; Effect of "circadian alignment condition": F=4.9; p=0.03). The relative abundance of the phylum Firmicutes decreased by −6.9 ± 2.1% from baseline CR to post-misalignment CR; whereas the relative abundances for Proteobacteria, Bacteroidetes and Fusobacteria increased by 8.3% ± 3.4%, 16.8% ± 5.6% and 16.5% ± 4.954%, respectively (Figure 3A). Significant misalignment effects were detected for the most prevalent genus taxa, i.e., *Streptococcus* (Firmicutes taxa; Prevalence: 27.77%; Effect of "circadian alignment condition": F=4; p=0.04). Furthermore, significant misalignment effects were also observed for two genera within the phylum Proteobacteria, *Haemophilus* (Prevalence: 9.4%; Effect of "circadian alignment condition": F=5; p=0.03) and *Neisseria* (Prevalence: 6.9%; Effect of "circadian alignment condition": F=14.4; p=0.002). Significant misalignment effects were detected for one genus within the phylum Fusobacteria, *Fusobacterium* (Prevalence: 4.8%; Effect of "circadian alignment condition": F=14.8; p=0.002). Significant misalignment effects also occurred for four genera within the phylum Bacteroidetes. These were *Prevotella* (Prevalence: 3.3%; Effect of "circadian alignment condition": F=13.6; p=0.004), *Porphyromonas* (Prevalence: 2.8%; Effect of "circadian alignment condition": F=18; p<0.001), *Gemellaceae Unclassified* (Prevalence: 2.3%; Effect of "circadian alignment condition": F=12.8; p=0.006) and *Capnocytophaga* (Prevalence: 1.3%; Effect of "circadian alignment condition": F=50.4; p<0.001). Collectively, we observed that the relative abundance of the most prevalent genus, *Streptococcus*, decreased by −5.5 ± 4.7% from baseline CR to post-misalignment CR, whereas the relative abundance of more minor taxa (those with an initial relative abundance ˂ 10%), increased from baseline CR to post-misalignment CR. The increases occurred for the taxa *Haemophilus* (by 7.3% ± 5.2%), *Neisseria* (8.2% ± 6.3%), *Fusobacterium* (22.6% ± 5.5%), *Prevotella* (58.4% ± 5.1%), *Gemellaceae* (62.1% ± 8. 9%) and *Capnocytophaga* (42.9% ± 55.3%) (Figure 3B). Furthermore, beta-diversity analysis using ANOSIM indicated that the circadian misalignment effect on saliva microbial communities, across individuals and at the group level, was significant (p˂0.05) (Table S2). Differences in the abundances of bacterial genera across each individual's circadian rhythmic conditions were determined and then summarized by high or low abundance due to the effects of circadian misalignment (Table S3). Of note, there were no significant differences in abundance of bacterial genera between grouped circadian rhythmic conditions.

We also assessed the effects of circadian misalignment on the endogenous circadian rhythms of oral microbiota community composition. No significant interaction of circadian alignment condition and circadian effect on the relative abundance of oral microbiota phyla and genera were observed, although some microbiota phyla/genera appeared to exhibit

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decreased circadian amplitude and seemed to lose rhythmicity after circadian misalignment (Figure S1 depicts the phylum Proteobacteria (**A**) and the genus *Neisseria* (**B**), genus *Fusobacterium* (**C**) and the genus *Streptococcus* (**D**) represented in Figure 2). Lastly, we assessed circadian misalignment effect on oral microbiota diversity (alpha diversity). We observed no significant change in alpha-diversity indices from CR1 to CR2 when the individuals were grouped. However, we did find significant differences in alpha diversity indices ($p < 0.05$) from CR1 to CR2 within five of the six individuals (Table S4), with an overall increase in all the alpha-diversity metrics from baseline CR1 to post-misalignment CR2.

Circadian misalignment impacts the functional pathways associated with oral microbiota populations

Lastly, we investigated the effects of circadian misalignment on the metagenome functional content of the oral microbiota (inferred gene content analysis). The relative abundance of fifteen functional pathways significantly changed ($p < 0.05$) from baseline CR to post-misalignment CR (Table 1). Circadian misalignment reduced the relative abundance of five functional pathways: phenylalanine, tyrosine and tryptophan biosynthesis; protein export; glycerolipid metabolism; replication, recombination and repair proteins; and thiamine metabolism. Conversely, circadian misalignment increased the relative abundance of ten functional pathways. These included sulfur metabolism; peroxisomes involved in catabolism of very long chain fatty acids; Peroxisome Proliferator-Activated Receptors (PPARs; nuclear hormone receptors activated by fatty acids); vibrio cholerae pathogenic cycle; lipoic acid metabolism; biosynthesis of vancomycin; cellular antigens; phenylpropanoid biosynthesis relevant to structural and signaling molecules; meiosis; and stilbenoid, diarylheptanoid and gingerol biosynthesis (Table 1).

Discussion

In this proof-of-concept study, we describe that the human oral microbiota community structure exhibits endogenous circadian rhythmicity, and that circadian misalignment dramatically alters the microbiota landscape, as well as inferred specific functional pathways associated with oral microbiota communities. While many studies have shown daily saliva and/or stool microbiota profiles in rodents (21, 23) and in humans (17, 24–26), to our knowledge, none has provided evidence of an endogenous circadian rhythm in any mammalian system. In other words, it remained to be established whether such rhythms persist in the absence of a rhythm in food intake and sleep, as well as in activity, light, and posture like in our CR conditions. Hence, preclinical work and the few human studies conducted thus so far have elucidated diurnal - *not circadian* - rhythms.

Previous work on the temporal fluctuations in the intestinal microbiota assessed over a long time frame (sampled for up to 5 years) showed remarkable stability of microbiota composition over time (48). However, emerging preclinical work using a finer temporal resolution (hours-scale) indicates that the intestinal microbiota exhibits diurnal oscillations under the control of host eating time, and that ablation of the host molecular clock *Period* (*Per)* genes - key components of the host molecular clock - causes dysbiosis (21, 49,

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50). Endogenous circadian rhythms in human intestinal microbiota have not yet been established due to the need for stringently controlled circadian paradigms that unmask the true expression of endogenous circadian rhythms (30, 31). In humans, analysis of stool is not ideal for such studies because it has slow temporal dynamics that depend on e.g., gastrointestinal motility/transit and represents an integrated measure over many hours (~8–72h of the gut biology) (51).

Currently, there is very limited evidence for temporal dynamics in gut microbiome in humans, with one pilot study having investigated rectal microbiome at 12 hourly intervals for 48 hours in real-life masked conditions (26). Given that saliva samples can be obtained with a higher frequency as compared to stool/rectal samples, it is possible to derive finer temporal dynamics for the oral microbial community composition and function. Oral and rectal microbiomes are likely to share some congruent taxa composition (26), although there are also differences in the composition and temporal changes between oral and gut (stool) microbiomes (52). Moreover, the oral cavity may serve as a reservoir for potential intestinal pathobionts associated with chronic inflammatory conditions that have the potential to further aggravate intestinal disease (53). Therefore, insights on the oral microbiome may provide important inferences on the gut microbiome. Daily rhythms in the oral microbial community composition and microbiota diversity exist in humans, which may be associated with metabolism in the host (24, 25). While it is still unclear whether the oral (saliva) microbiota displays intrinsic clocks or circadian rhythms, they very likely do, given that the oral and gut microbiota compositions share some congruent taxa composition. Currently, there is emerging evidence that some gut (stool) microbiota display intrinsic circadian clocks and circadian rhythms, and that the gut microbiota can be affected by the host circadian signals, such as melatonin (54) and by changes in the host circadian rhythms due to e.g., food composition and timing (25, 55). Of note, circadian oscillations have been shown to be completely abolished by incubating saliva in vitro, thus suggesting that host physiological changes mostly contributed to the microbial oscillation (24). Here, we go beyond diurnal oscillations in the oral microbiota composition and provide evidence for endogenous circadian rhythms. However, we observed that effects of circadian rhythms in (i.e., Metacycle ARSER), and circadian misalignment effects (i.e., alpha- and beta-diversity) on, the structure of oral microbial communities varied strongly across individuals. Notwithstanding the variability amongst individuals, clear group-based rhythms and circadian misalignment effects in beta diversity were apparent. The cosinor grouped-based analyses show endogenous circadian rhythms in the oral microbiota phylum Proteobacteria, which contains many proinflammatory taxa (56), and the genera *Streptococcus*, *Neisseria* and *Fusobacterium*, involved in immune control (57). Interestingly, circadian rhythms in the current study occurred in ~20% of oral microbial genera, strikingly akin to the ~20% of intestinal (stool) microbiota in mice that exhibit diurnal fluctuations in their relative abundance (21, 23), suggesting that part of the orogastrointestinal microbiota displays endogenous circadian oscillations.

Contemporary lifestyle leads to behaviors and environments at odds with our endogenous circadian rhythms, resulting in circadian misalignment, a biological phenomenon often experienced by shift workers and frequent flyers across time zones (jet lag). In our study, the first CR protocol (baseline CR) allowed establishing the existence of endogenous circadian

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rhythms in oral microbiota composition, as i. circadian rhythms were inferred without masking effects, and ii. participants were under normally aligned circadian conditions up until then. The second CR protocol (post-misalignment CR) allowed inferring i. the effects of prior circadian misalignment on oral microbiota composition and ii. the effects of prior circadian misalignment on the endogenous circadian rhythms of the oral microbiota composition. As described above, we show that there are rhythms that persist even in the absence of any behavioral and environmental rhythms, which is of great importance for individuals experiencing circadian misalignment. For instance, if microbiota population rhythms were only driven by behavioral and environmental rhythms, then they would easily and rapidly follow shifts in daily behavioral/environmental rhythms as in shift workers, without adverse consequences. However, if these microbiota rhythms are driven at least partially independently of behaviors and/or environment (i.e., by the endogenous circadian system), this may explain why circadian misalignment can disrupt microbiota population structure. In support of that, we find that even relatively short-term experimental circadian misalignment induces dramatic changes in the relative abundance of human oral microbiota populations. These microbial alterations are indicated by decreases in the relative abundance of the most prevalent phylum Firmicutes and genus *Streptococcus* (i.e., primary colonizing putative beneficial stabilizing oral bacteria of the mouth), plus increases in the relative abundance of the less prevalent phylum Bacteroidetes and four putative pro-inflammatory pathogenic oral genera within this phylum (58–60). Genera such as *Fusobacteria* and *Porphyromonas*, whose relative abundance increased following misalignment, are commonly associated with pro-inflammatory states, such as periodontal diseases and oral and colon cancer (61, 62). In the long-term, frequent exposure to circadian misalignment might result in a microbiota-driven heightened proinflammatory state with ramifications for health and disease. Moreover, as shown in Figure S1, we observed no significant interaction of circadian alignment condition and circadian effect on the relative abundance of oral microbiota phyla and genera, although some microbiota phyla/genera appeared to exhibit decreased circadian amplitude and seemed to lose rhythmicity after circadian misalignment. As such, it is possible that exposure to circadian misalignment might also adversely affect the endogenous circadian rhythms in oral microbiota composition. However, these latter effects require replication in adequately powered human laboratory studies.

Prior animal experimental work has shown that both environmental and genetic circadian rhythm disruption perturb microbiota communities and cause intestinal hyperpermeability (18, 19, 23). For instance, animal work suggests that environmental circadian disruption (a simulated jet lag model (that had a shift of the light/dark cycle by 8h), changes microbiota composition (21). Furthermore, germ-free mice colonized through fecal transfer with the microbiota derived from individuals experiencing jet lag had increased weight gain and glucose levels following an oral glucose challenge, as compared to mice that received microbiota from the same individuals prior to jet lag (21). Jointly, these findings highlight the impact of circadian disruption on the microbiota and their functional relevance for metabolic control. Further studies including host transcriptomics, microbial metagenomics and metatranscriptomic sequencing, and metabolite analyses will be required to understand the role of oral microbiota in circadian misalignment effects. Given the magnitude of the circadian misalignment effects on the oral microbiota populations, one may infer that

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these changes can influence the abundance and expression of genes in functional pathways associated with these microbiota populations. Here, we show through inferred gene content analysis that circadian misalignment decreased phenylalanine, tyrosine, thiamine and tryptophan pathways, which mediate amino acid metabolism and immunity (63). Conversely, exposure to circadian misalignment increased the sulfur biosynthesis pathway, which controls the translation of specific genes, with potential pro-carcinogen implications when this pathway is disrupted (64). The host circadian misalignment affected bacterial functional gene content with increased abundances in lipoic acid metabolism and cellular antigen pathways. These inferred functional pathways have been associated with systemic inflammation (65, 66), wherein the increased abundances in both peroxisomes and PPAR signalling modulated systemic oxidative stress and inflammation (67, 68). Because we used PICRUSt on 16S rRNA gene amplicon sequencing data, we can only infer the effect of host circadian misalignment on oral microbiota functional gene content and functional pathways. Therefore, the functional relevance of our reported findings on oral microbiota function should be interpreted cautiously. Further studies using e.g., metagenomic shotgun sequencing are required for the direct assessment of oral microbiota functional gene content.

Our reported findings are unlikely to have been modulated by underlying health conditions, including sleep disorders. While no clinical PSG or home sleep apnea test were performed as part of the screening process, participants were relatively young (18–35 y) with a relatively low BMI (18–28 kg/m^2). Moreover, participants were free from any acute, chronic or debilitating medical conditions based on clinical history, electrocardiogram, clinical biochemical screening tests of blood and urine, and a physical examination by a physician, prior to the study. Among the list of illnesses that would have resulted in study exclusion (verified through questionnaires and during physical examination) was a history of sleep disorders. Moreover, we observed no significant differences in sleep structure characteristics between the circadian alignment and misalignment conditions under the FD protocol (Table S5). Therefore, subtle sleep structure differences are unlikely to modulate our reported findings on oral microbiota composition.

Strengths to our study include our stringent inclusion/exclusion criteria that ruled out potential confounders, including comorbidities and medication intake, and the stringent circadian laboratory protocol that allowed determining endogenous circadian rhythms in oral microbiota, and the effects of circadian misalignment. Limitations of our study include sample size, which was partly due to our stringent participant inclusion criteria and the demanding nature of the laboratory protocols. Because our study is likely underpowered, we could not perform statistical corrections for multiple comparisons. As such, it is essential for these findings to be replicated in a larger trial with adequate statistical power. Our cosinor analyses included seven time point measurements during the baseline CR and nine time point measurements during the post-misalignment CR. Because of that, part of the circadian misalignment effects may be due to time in the study protocol. Of note, previous human Forced Desynchrony protocols (10, 69) have shown that circadian misalignment effects on e.g., metabolic outcomes are reversible, suggesting that time duration during the laboratory studies may not explain the circadian misalignment effects. However, those laboratory protocols and the outcome measures (e.g., glucose tolerance in those studies) are rather different from our study. Therefore, future studies are needed to investigate if the

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observed misalignment effects on oral microbiota taxa composition recover after circadian realignment. Collectively, these findings suggest that circadian misalignment may adversely affect specific functional pathways integral to metabolic and immune control. However, our data do not provide an indication of altered expression levels, as analyses are based on microbiota relative abundances. Therefore, future studies are needed to examine altered host and microbial gene expression patterns under circadian misalignment.

Conclusion

Human health intertwines with the integrity of microbiota composition and function. While habitual diet, changes in macronutrient composition, timing of food consumption, and the ingestion of non-digestible dietary fibers have been shown to impact the human microbiota (25, 70, 71), it is unclear if circadian misalignment *per se* (typical in night shift work and jet lag) impacts human microbiota. Collectively, our proof-of-concept study provides evidence for human oral microbiota circadian rhythmicity and show that circadian misalignment can rapidly and dramatically modify human microbial population structure and functional pathways involved in metabolism and immunity. These findings have translational relevance to individuals typically exposed to circadian misalignment, including night shift workers and frequent flyers.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Nonstandard abbreviations list

ANOSIM	Analysis of similarity
ARSER	Autoregressive spectral estimation regression
BIOM	Biological observation matrix
BMI	Body-mass index
CBT	Core body temperature
CR	Constant routine
FD	Forced desynchrony

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FDR	False-discovery rate
KEGG	Kyoto Encyclopedia of Genes and Genomes
OTU	Operational taxonomic units
PCR	Polymerase chain reaction
PICRUSt	Phylogenetic Investigation of Communities by Reconstruction of Unobserved States
PPAR	Peroxisome Proliferator-Activated Receptor
T2DM	Type 2 Diabetes Mellitus

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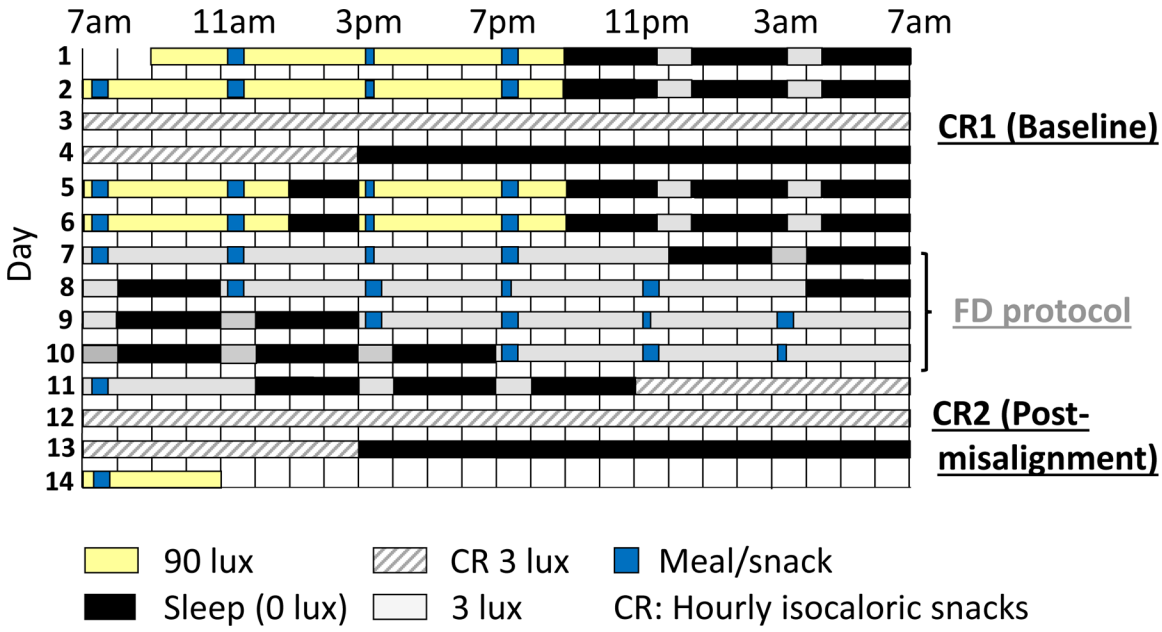


Figure 1. Study design.

The 14-day in-laboratory study design is presented with relative clock time (for an example research participant with a habitual wake-up time of 7:00AM) along the X-axis and study day along the Y-axis. Data presented in the manuscript are derived from the two constant routine protocols, Baseline CR (CR1; days 3–4) and Post-misalignment CR (CR2, days 11–13). During CR protocols, participants were kept awake, under constant posture (semi-recumbent, upper body at 45° angle), constant dim light (light levels <3lx), constant room temperature and humidity, and received isocaloric snacks every hour. Oral microbial community composition was assessed through saliva sampling every 4-h during CR conditions.

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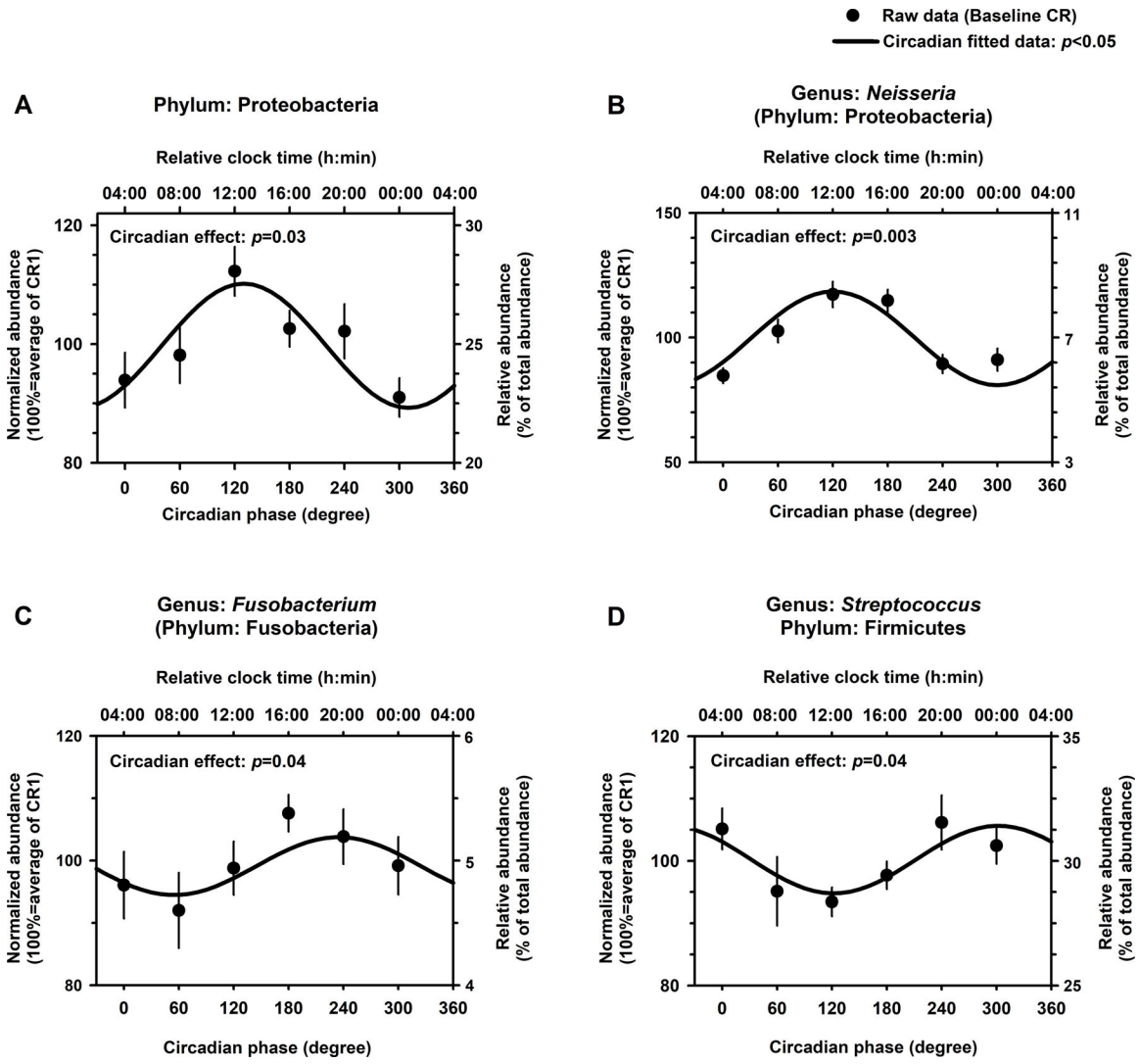


Figure 2. Endogenous circadian rhythms in human oral microbiota.

Cosinor mixed-models of circadian rhythms were applied to the oral microbiota community composition. Phylum Proteobacteria (**A**) and the genus *Neisseria* (**B**) had significant endogenous circadian rhythms (Baseline CR). Genus *Fusobacterium* (**C**) and the genus *Streptococcus* (**D**) showed significant endogenous circadian rhythms (Baseline CR). The cosinor models use precise circadian phase data. To show that these models adequately fit the actual data, we also plot average data grouped into 60-circadian degree windows (~4-hour resolution) with standard error of the mean (SEM) error bars. Bottom x-axes corresponds to circadian phase with 0° indicating the timing of the fitted circadian core body temperature minimum (average ~3:30AM during baseline CR in these participants). Top x-axes indicate the corresponding average clock time in these participants; left y-axes correspond to the percentage of each individual's mean during Baseline CR; right y-axes indicate the relative abundance (% of total abundance); *P* values: significance of "circadian effect" derived from the cosinor mixed-model analyses.

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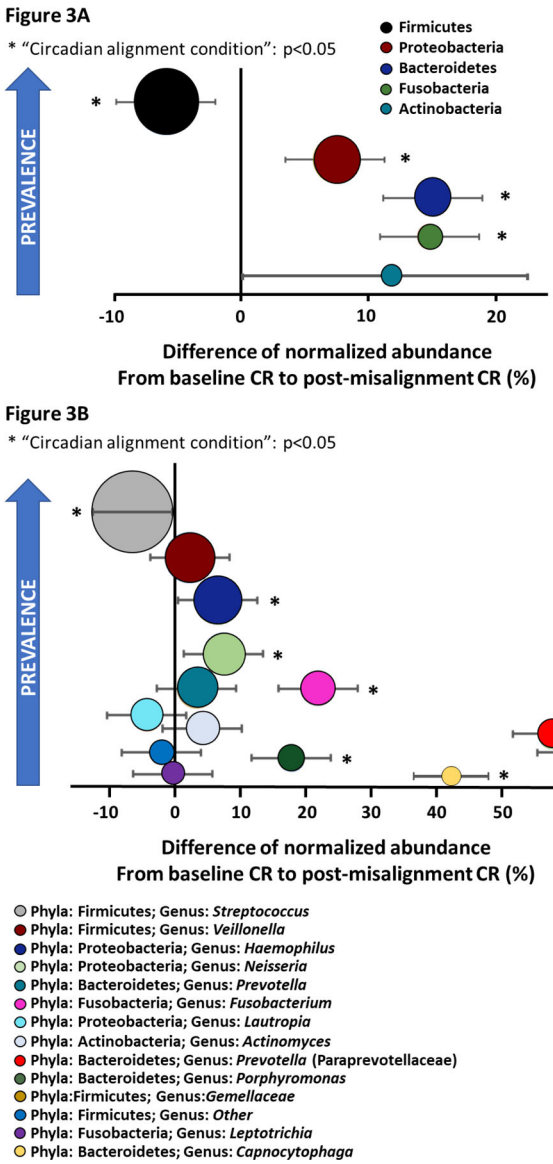


Figure 3. Circadian misalignment affects human oral microbiota.

Bubble plots graphically illustrate the circadian misalignment effects on oral microbiota composition at the phyla (A) and genera (B) taxonomical levels. Data correspond to the five dominant phyla (>1% threshold of relative abundance) and to the 14 dominant genera (>1% threshold of relative abundance). Each bubble plot corresponds to the difference from the baseline CR to the post-misalignment CR (CR2-CR1). X-axes: data was i. normalized as 100% of each participant's mean during Baseline CR1, ii. detrended to remove the linear component, iii. averaged across all time points for each CR condition, iv. derived the difference from the baseline CR to post-misalignment CR at the individual level, and v. averaged across all participants, per phylum and genus. Y-axes: 5 dominant phyla (A) and 14 dominant genera (B), ordered by prevalence. The area of each bubble plot corresponds to the prevalence for each given phylum and genus averaged across all participants in both CR protocols. Data correspond to mean and standard error of the mean (SEM); P-values:

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significance of “circadian alignment condition” (aligned vs. misaligned) is derived from the cosinor mixed-model analyses (see “Statistical analyses“ for details).

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Table 1.

Effects of post-circadian misalignment on abundances of inferred microbiota functional pathways.

Functional pathway	Change in abundance due to Circadian Misalignment	Average Difference of Abundance between Post-Misalignment CR2 and Baseline CR1 * Abundance Mean ± (SEM)	*P*-Value #	Prevalence % **	% Change from Baseline CR1 ***
Phenylalanine, tyrosine and tryptophan biosynthesis	Decreased	−5,397.90 ± (3,524.92)	0.027	0.84	−2.48
Replication, recombination and repair proteins	Decreased	−4,934.09 ± (3,621.85)	0.040	0.81	−2.35
Protein export	Decreased	−3,680.56 ± (2,400.31)	0.042	0.74	−1.92
Thiamine metabolism	Decreased	−3,463.62 ± (1,646.66)	0.034	0.47	−2.81
Glycerolipid metabolism	Decreased	−3,406.85± (2,243.37)	0.028	0.38	−3.46
Peroxisome	Increased	784.67 ± (922.51)	0.038	0.18	1.69
Sulfur metabolism	Increased	751.70 ± (997.90)	0.048	0.25	1.17
PPAR signaling pathway	Increased	702.34 ± (659.52)	0.038	0.10	2.67
Lipoic acid metabolism	Increased	672.12 ± (592.19)	0.015	0.07	3.91
Cellular antigens	Increased	537.17 ± (383.41)	0.016	0.05	4.59
Meiosis - yeast	Increased	399.39 ± (125.18)	0.003	0.01	20.62
Phenylpropanoid biosynthesis	Increased	384.23 ± (352.34)	0.005	0.04	3.52
Vibrio cholerae pathogenic cycle	Increased	235.34 ± (430.36)	0.020	0.08	1.13
Stilbenoid, diarylheptanoid and gingerol biosynthesis	Increased	215.13 ± (1.09)	0.0009	0.01	13.79
Biosynthesis of vancomycin group antibiotics	Increased	62.91 ± (216.30)	0.040	0.06	0.42

(*)= Average of the difference between the circadian misalignment CR2 abundance and baseline CR1 abundance [average (CR2-CR1)], across all subjects.

(#) *P*-Values (p˂0.05) significant differential abundances between the two circadian conditions determined using the software program STAMP.

(**)= Prevalence was determined using the abundance of each pathway divided by the sum total of the entire dataset (in %) of each functional pathway.

(***)= Percentage of [average (CR2-C1) / average (CR1)], across all subjects. (SEM) = Standard error of the mean. SEMs are increased due to variability within and between subjects. PICRUSt abundance data was not rarefied but rather filtered at 0.01% threshold. Any pathway that had an overall abundance of at least 0.01% of the total abundance was retained for analysis.